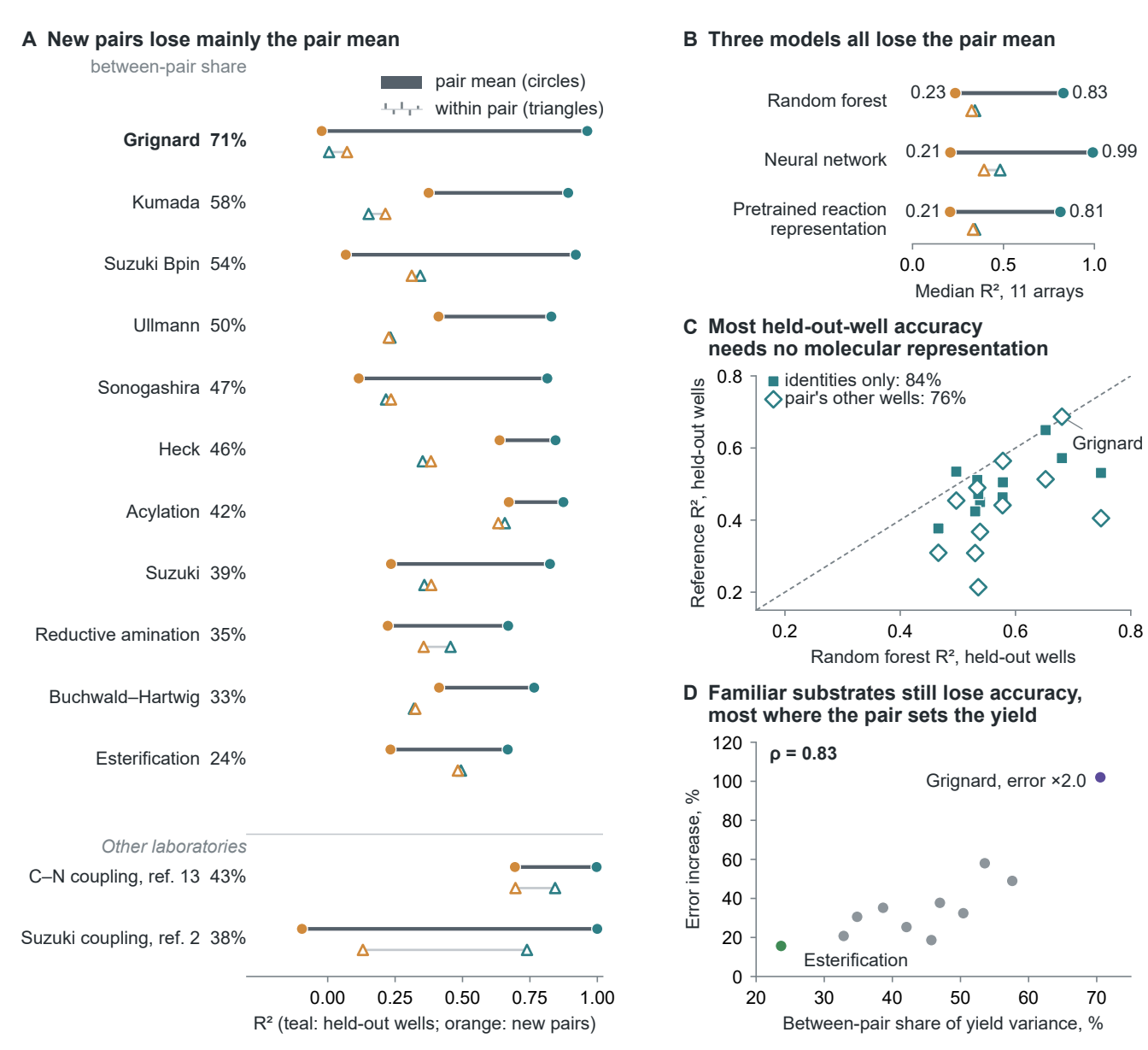

A New pairs lose mainly the pair mean
between-pair share
pair mean (circles)
within pair (triangles)
Grignard 71%
Kumada 58%
Suzuki Bpin 54%
Ullmann 50%
Sonogashira 47%
Heck 46%
Acylation 42%
Suzuki 39%
Reductive amination 35%
Buchwald–Hartwig 33%
Esterification 24%
Other laboratories
C–N coupling, ref. 13 43%
Suzuki coupling, ref. 2 38%
0.00 0.25 0.50 0.75 1.00
R² (teal: held-out wells; orange: new pairs)
B Three models all lose the pair mean
Random forest 0.23 0.83
Neural network 0.21 0.99
Pretrained reaction representation 0.21 0.81
0.0 0.5 1.0
Median R², 11 arrays
C Most held-out-well accuracy needs no molecular representation
identities only: 84%
pair's other wells: 76%
Grignard
Reference R², held-out wells
0.8 0.6 0.4 0.2
0.2 0.4 0.6 0.8
Random forest R², held-out wells
D Familiar substrates still lose accuracy, most where the pair sets the yield
ρ = 0.83
Grignard, error ×2.0
Esterification
Error increase, %
120 100 80 60 40 20 0
20 30 40 50 60 70
Between-pair share of yield variance, %